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Cluster Based Sleep Transistor Approach for Low Power 6T SRAM Cell

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Abstract
The demand of high density VLSI circuits has been increasing as the size of ICs is becoming small day by day. It has been seen that, to reduce active power dissipation in memory, the supply voltage reduction is necessary. Lowering the supply voltage is one of the most straightforward and effective ways to suppress the energy consumption because reducing the supply voltage could reduce the dynamic power and leakage power remarkably. The static power dissipation occurs in an idle mode of the circuit. In this paper, the analysis has been carried out on simple 6T SRAM cell using NMOS sleep transistor and current flowing through the sleep transistor in active and sleep mode has been taken out as well as the effect of changing the width of sleep transistor has been compared. In this work, a technique called 'clustering technique' has been proposed to reduce the active power requirement and the simulation has been done on 4X1 SRAM cell. In this work first of all, the power dissipation of 4X1 SRAM cell without sleep is taken out. Then, the clustering technique is applied in which, first of all the cells are connected to an individual sleep then the sleep is shared between the two cells and after that all the four cells are connected to only one sleep. Therefore, the result with clustering technique shows an improved result than an individual sleep when connected to the SRAM cell. In this work, the SRAM cell without sleep transistor dissipates more power during different states as compared to SRAM cell with an individual sleep transistor. SRAM cell with sleep dissipates 26.65 % less power during write '1' operation, 59.34 % less power during hold operation and 16.74 % less power during read operation. In clustering technique it has been further reduced, during write '1' operation the cell dissipates 49.09 % less power, while write '0' operation it consumes 66.29 % less power, during read operation it dissipates 42.67 % less power as compared to the SRAM cell without sleep transistor.

Keywords: *VLSI circuits, SRAM, NMOS, MTCMOS, 4X1 SRAM cell*

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INTRODUCTION

As with every generation of technology, the demand of handling the large amount of data in embedded memory has been increasing. To fulfill the requirement of handling large data feature size of transistor is continuously reducing. With respect to high transistor density the problem of power consumption is becoming a prominent issue to tackle [1, 2]. Static random access memory is the first choice of designing semiconductor embedded memories because of low power dissipation [3]. The low-power feature for on-chip SRAMs is becoming more important, especially for battery-operated portable applications. It is however, one of the most significant challenges of high-density VLSI circuit [4, 5]. The main objective of this paper is to estimate the effect of clustering technique on 6T SRAM cell and to investigate transistor sizing of the 6T SRAM cell for optimum power and delay as shown in the Table 1. In this work, an average power dissipation of 6T SRAM cell has been compared with SRAM cell using clustering technique. The clustering technique reduces the power dissipation of 6T SRAM cell in read, write and hold operation.

CONVENTIONAL 6T SRAMCELL

The schematic diagram of conventional 6T SRAM cell is shown in the below Figure 1. This memory cell has two inverters connected back to back. There are two access transistors (N3 and N4), which are controlled by the word line (wl) [2]. The cell can retain its one of the two possible states "0" or "1".

There are mainly three states of SRAM memory cell, read, write and hold states [4].

(i) During read operation, WL voltage is raised high, bl and blbr were on the same potential. The memory cell discharges either bl (bit line) or blbr (bit line complement), depending on the stored data on nodes out 1 and out 2 [6].
(ii) During write operation, the WL is kept high, bl and blbr should be complement to each other. The value to be written is applied to the bit lines [7].
(ii) In the hold state of memory cell the word line is kept low and the bit lines can be at any of the one state 0/1, because these are disconnected from the memory cell [8].The two cross-coupled inverters P_1, N_2 and P_2, N_1 shown in Figure 1, will try to flip the outputs of the out 1 and out 2 since, they are totally isolated from the outside world [7].

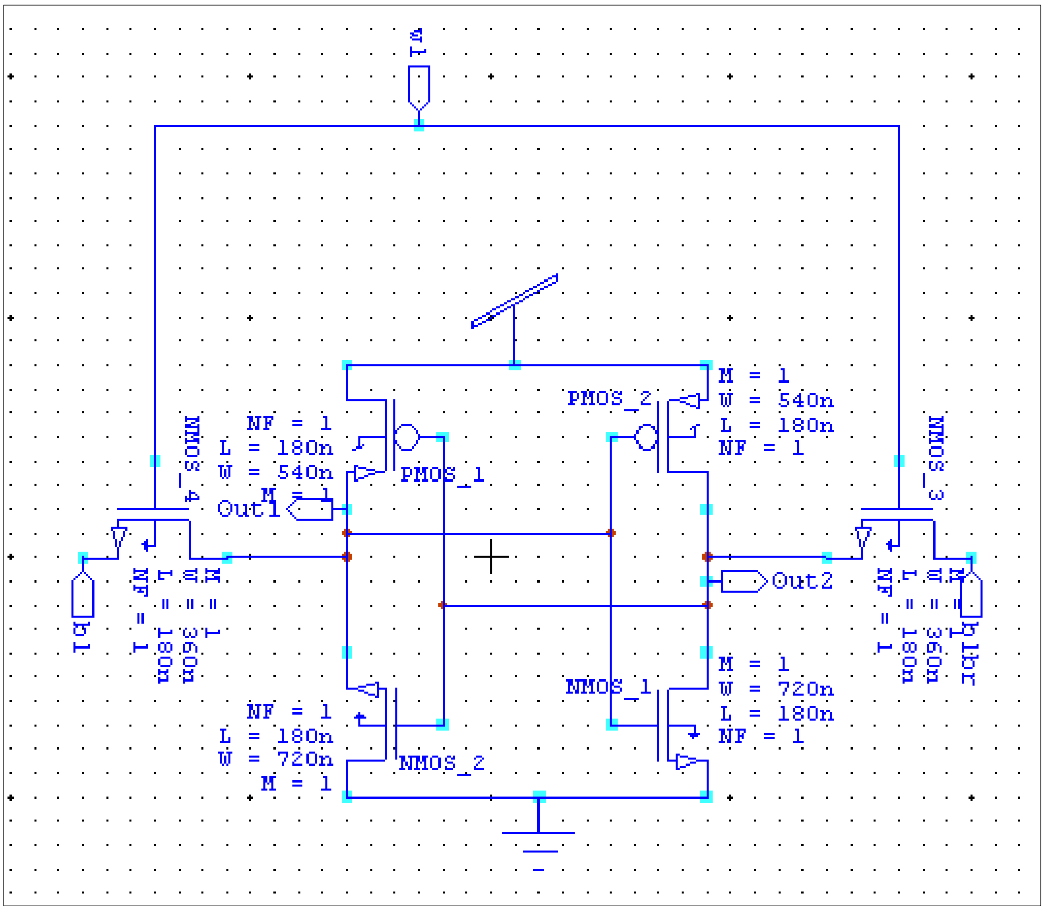

Fig. 1: *Conventional 6T SRAM Cell.*

SRAM CELL WITH SLEEP TRANSISTOR

To see the effect of sleep transistor on 6T SRAM cell, NMOS sleep transistor is connected between the 6T SRAM cell and ground as shown in the below Figure 2. During an active mode the NMOS sleep transistor is ON, which increases the speed of the circuit. During standby mode the sleep transistor is OFF, which reduces the leakage current [9]. In this circuit, the effect of W/L ratio on the sleep transistor is observed for the leakage current flowing through the sleep transistor in active and standby mode.

Table 1 shows the leakage current flowing through the sleep transistor and voltage at the drain node of sleep transistor in active and sleep mode. It can be seen that the current flowing through sleep transistor increases with respect to the width of the transistor in active and standby mode. Whereas, the voltage at drain node of the sleep transistor decreases while increasing the width of that transistor, which is responsible for more and less leakage current flowing through the transistor. As the width of the transistor increases the propagation delay of the SRAM cell increases as shown in Table 1.

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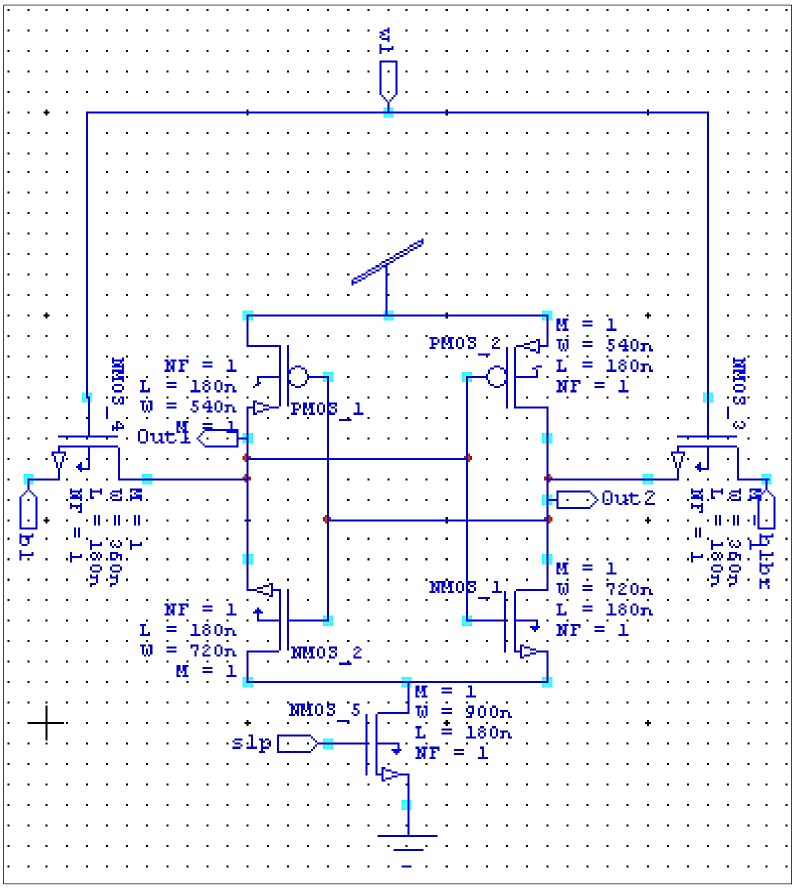

Fig. 2: *SRAM Cell with Sleep Transistor.*

Table 1: *Comparison of Delay, Current Flowing Through the Sleep Transistor at Different W/L Ratio of Sleep Transistor.*

W/L ratio (nm)	Delay in (ps)	I_D current through sleep transistor		Voltage at the drain node of sleep transistor	
		Active mode	Sleep mode	Active mode	Sleep mode
360/180	50.170	18.038p	7.022a	2.034e-008	5.361e-008
720/180	51.885	20.93p	10.66a	1.500e-008	5.361e-008
1080/180	52.521	22.715p	14.22a	1.171e-008	5.361e-008
1440/180	52.932	23.861p	17.73a	9.605e-009	5.361e-008
1800/180	53.377	24.65p	21.23a	8.136e-009	5.361e-008

4X1 CELL WITHOUT SLEEP TRANSISTOR

It describes the designing of 4X1 SRAM cell arrays of 1 row and 4 columns. Each block of the array is of 6T SRAM cell. There is 1 row and 4 columns that are arranged to form a 4X1 SRAM cell array. There is only one row, so there is no need of decoder. All the 4 cells are connected to the same word line (wl). As the row consists of 4 cells, it constitutes to form half a byte [3]. The data is written into the cell through multiplexers which are connected to the bit lines. To read the data stored on storing nodes out 1 and out 2, sense amplifiers are connected which read the stored values of the SRAM cell. The average power dissipation during read, write and hold states are shown in Table 2. The 4X1 cell array without any sleep transistor shows the highest power dissipation during all the states (read, write and hold state). The average power dissipation during read, write and hold states are shown in Table 2. The 4X1 cells array without any sleep transistor shows highest power dissipation during all the states (read, write and hold state). The 4X1 SRAM cell without any sleep transistor is shown in the below Figure 3.

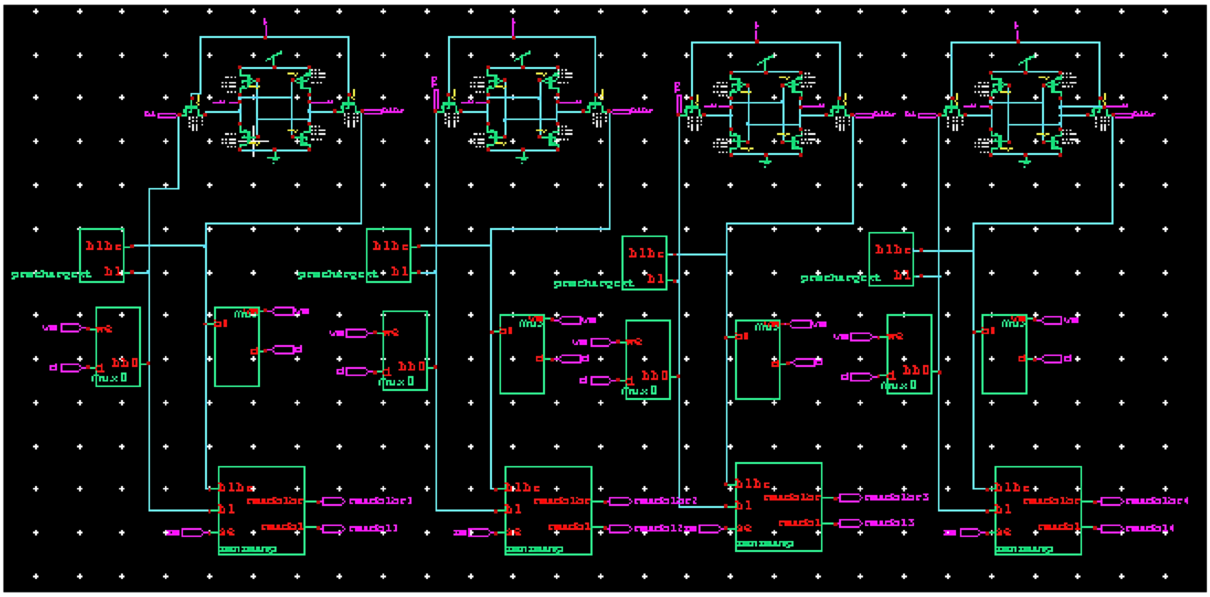

Fig. 3: *4X1 Cell without Sleep Transistor.*

4X1 CELL WITH INDIVIDUAL SLEEP TRANSISTOR

Figure 4 shows, the 4X1 cell with individual sleep transistor. The W/L ratio of the sleep transistor is 900/180 nm, which is individually connected to four cells. When all the four cells are individually connected to the sleep transistors, the average power dissipation during all the states increases as compared to the 4X1 cell without sleep transistors [10–12].

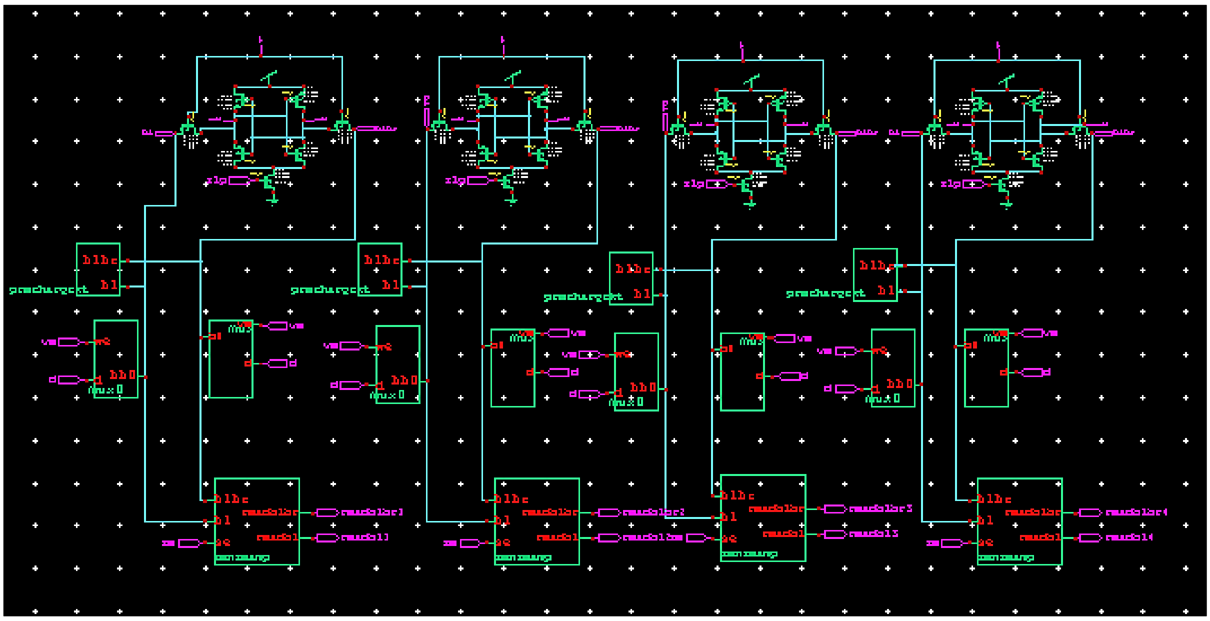

Fig. 4: *4x1 Cell with Individual Sleep Transistor.*

PROPOSED WORK

In this proposed work, the clustering technique has been used. In this technique, 6T SRAM cell with sleep transistor is used. As there are 4 cells in 4X1 cells array, each cell has got its individual sleep transistor. But, in the clustering technique instead of having individual sleep transistor, two or all cells can be connected to the same sleep transistor. This effects the average power dissipation of the memory cell during different modes of operation such as read, write and hold states.

4X1 Cell with Common Sleep Transistor

Figure 5 shows, the 4X1 cells array with two cells having a common sleep transistor. In between four cells there are two sleep transistors. Figure 6 shows the 4X1 cells array where all the four cells are sharing a single sleep transistor. As in this type of clustering, all the cells are connected with a common sleep, the number of transistor are reducing as well as it reduces the power dissipation during read, write and hold state of the memory cell as compared to the cells without sleep and cells with individual sleep. Comparison of all the techniques has been carried out in Table 2.

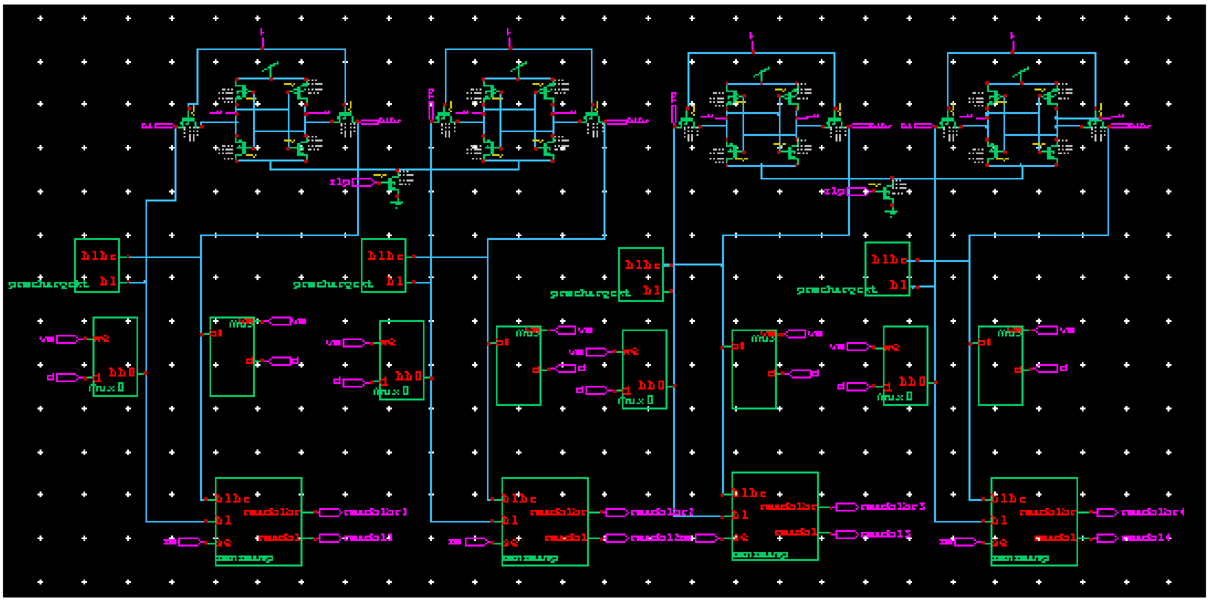

Fig. 5: *Two Cells with Common Sleep in 4X1 SRAM Cell.*

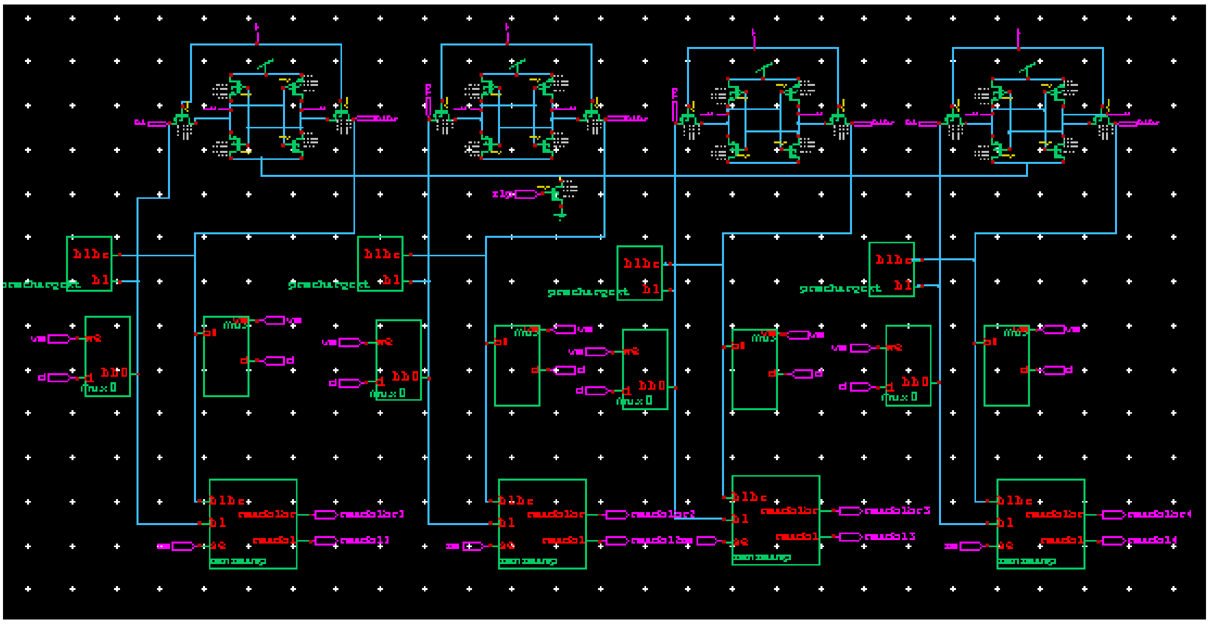

Fig. 6: *Four Cells with Common Sleep in 4X1 SRAM Cell.*

RESULTS

We estimated the impact of clustering technique on average power dissipation of 4X1 SRAM cell array [13]. The circuit is characterized by using the180 nm technology with the supply voltage of 1.8 v. All the simulations are done on tanner tool using TSMC spice file. The comparison of average power dissipation during write, read and hold operation for different combination of clustering technique is shown in Table 2. It can be easily seen that SRAM cell without sleep transistor dissipates more power during different states as compared to SRAM cell with individual sleep transistor. SRAM cell with sleep dissipates 26.65 % less power during write '1' operation, 59.34 % less power during hold operation and 16.74 % less power during read operation (Figures 7–10). This reduced power dissipation using sleep transistor has been reduced further by proposed technique (two cells are connected with common sleep and four cells are connected with common sleep). So, the proposed clustered based SRAM cell is low power efficient for all the modes of operation [14].

Table 2: *Comparison of Average Power Dissipation of 4X1 SRAM Cell During Read, Write and Hold State.*

Different modes of operation	Average power dissipation of SRAM cell Without sleep (in watts)	Average power dissipation of SRAM cells using Sleep Clustering Technique (in watts)		
		With individual sleep	Two cells with common sleep	Four cells with common sleep
Write "1"	2.6980e-003	1.9788e-003	1.374e-003	1.3734e-003
Write "0"	2.6992e-003	2.9570e-003	2.7667e-003	9.099e-004
Hold	1.1398e-003	4.6340e-004	4.6340e-004	4.6340e-004
Read	4.9660e-004	4.1344e-004	3.5612e-004	2.847e-004

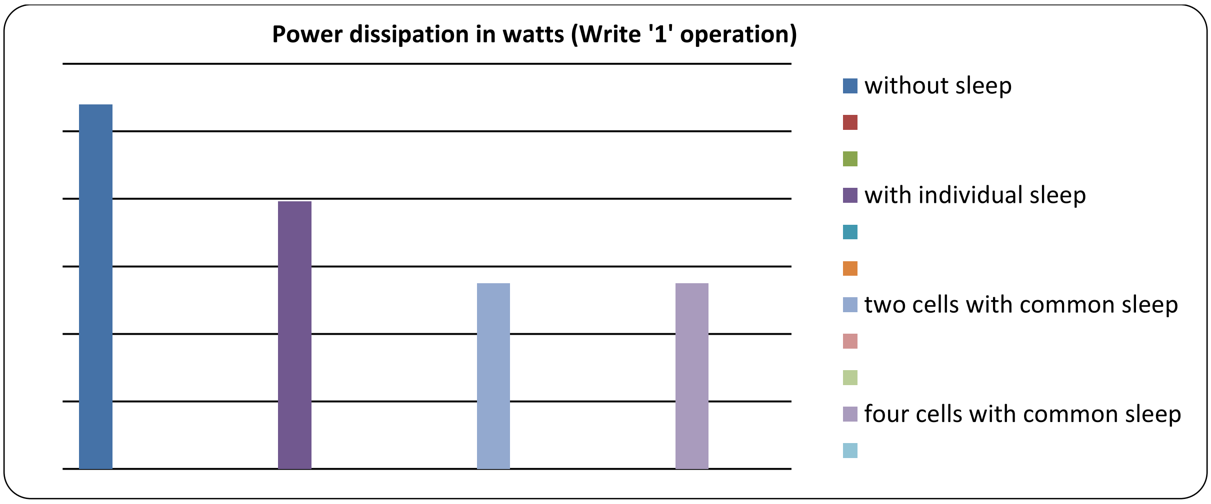

Fig. 7: *Comparison of Average Power Dissipation for Write "1" Operation.*

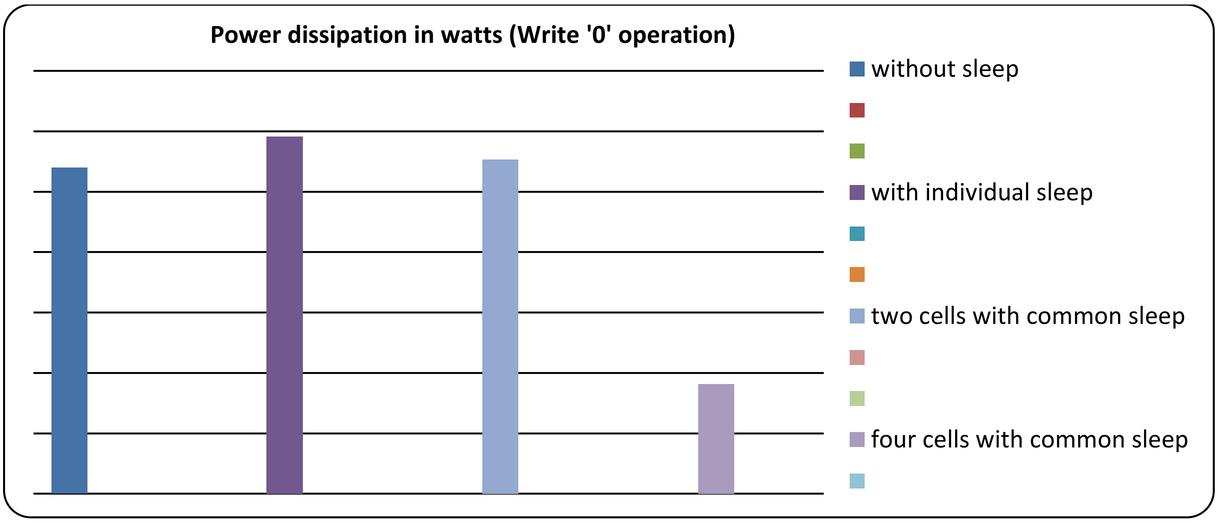

Fig. 8: *Comparison of Average Power Dissipation for Write "0" Operation.*

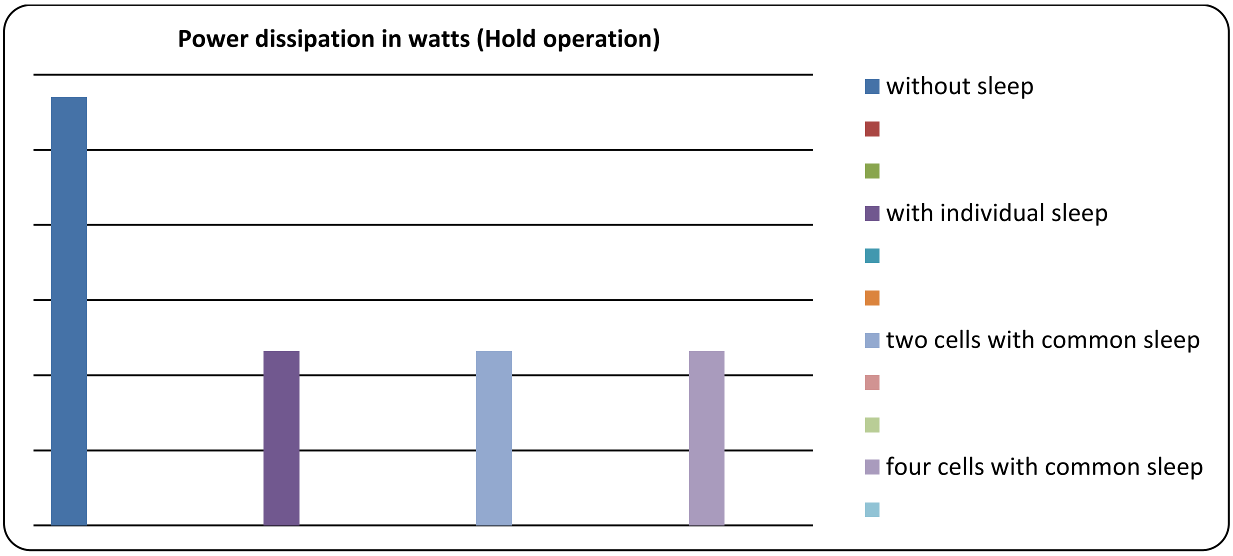

***Fig. 9:** Comparison of Average Power Dissipation during Hold Operation.*

***Fig. 10:** Comparison of Average Power Dissipation during Read Operation.*

CONCLUSION

This paper overall reveals that the SRAM cell with sleep transistor dissipates less power as compared to the cell without sleep transistor. SRAM cell with sleep transistor dissipates 26.65 % less power during write '1' operation, 59.34 % less power during hold operation and 16.74 % less power during read operation as compared to the cell without sleep transistor. This average power dissipation can be further reduced by applying clustering technique, in which rather connecting sleep transistor to each cell separately it can be shared by two cells or by all the four cells in 4X1 SRAM cells array. In clustering technique during write '1' operation the cell dissipates 49.09 % less power, while write '0' operation it consumes 66.29 % less power, during read operation it dissipates 42.67 % less power as compare to SRAM cell without sleep transistor. This technique results in less average power dissipates by the SRAM cell array during different states read, write and hold. As well as it reduces the area consumption taken by SRAM cells in ICs.

COMPETING INTERESTS

The authors declare that they have no competing interests.

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